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Unraveling V(V)-V(IV)-V(III)-V(II) Redox Electrochemistry in Highly Concentrated Mixed Acidic Media for a Vanadium Redox Flow Battery: Origin of the Parasitic Hydrogen Evolution Reaction

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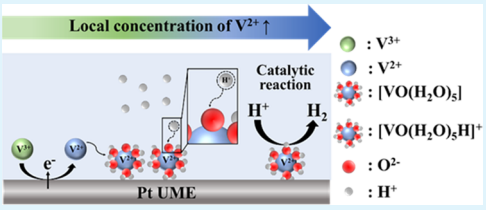
Supporting Information

ABSTRACT: We present a mechanistic understanding of the full redox electrochemistry of V(V)-V(IV)-V(III)-V(II) and the origin of the parasitic hydrogen evolution reaction (HER) during electroreduction of either V^{3+} or VO^{2+} in a highly concentrated mixed acidic solution based on both electroanalytical and computational approaches. First, we found that the VO^{2+}/VO_2^+ redox reaction is well explained by the EC/EC square scheme. We also found that V^{3+} is electrochemically oxidized to V^{4+} and subsequently undergoes a transition to stable VO^{2+} via hydrolysis. In the V^{3+}/V^{2+} redox reaction via voltammetric analysis at scan rates greater than 0.05 V/s, the voltammograms are well explained based on a simple $1e^-$ transfer reaction scheme. However, at the longer time scale observed in the chronoamperograms with constantly applied potentials where V^{3+} is electrochemically reduced to V^{2+}, we found that a significant HER occurs because of possible formation of an electrocatalyst related to the V(II)O species, $V(II)_{catalyst}$. We suggest that V(II)O is kinetically formed from V^{2+} via hydrolysis only when a local concentration of V^{2+} is high in the vicinity of a GC electrode surface, and V(II)O is adsorbed on a GC surface to form $V(II)_{catalyst}$. To extend our mechanistic pathway, electroreduction of VO^{2+} to V(II) was also analyzed, revealing that VO^{2+} is electroreduced to VO^+ and further reduced to VO in addition to disproportionation of VO^+. Eventually, $V(II)_{catalyst}$ forms on a GC electrode, resulting in a significant HER. The computational calculation strongly supports the possible formation of $V(II)_{catalyst}$. The calculation shows that neither V^{3+} nor V^{2+} can form stable intermediates during the HER, while V(II)O has the highest proton affinity compared with $V(III)O^+$ and $V(IV)O^{2+}$, indicating a plausible electrocatalytic property of V(II)O for the HER.



KEYWORDS: *all vanadium redox mechanism, hydrogen evolution reaction, vanadium(II) oxide, electrocatalyst, vanadium redox flow battery*

INTRODUCTION

Grid-scale electrical energy storage (EES) has been considered a powerful solution for the inherent electricity fluctuation generated from renewable energy sources, such as solar and wind power.[1] Among the suitable candidates for EESs, the redox flow battery (RFB)[2−4] has attracted significant attention due to its long cycle life and safety; most of RFBs are composed of aqueous electrolytes, which can be applied in a manner that avoids possible catastrophic conflagration. RFB is composed of two electrolytes dissolving different redox species, which are continuously circulated from the reservoirs to the electrochemical cell, with the anode and cathode physically separated by an ion-exchange membrane. Thermodynamic cell potential is determined by each equilibrium potential of redox species, which contain the anolyte and the catholyte, and energy density is directly proportional to concentration of dissolved redox species and electron numbers in the total redox reaction in the charge and discharge process.[5,6]

An all-vanadium RFB (VRFB),[6−11] which was first developed by Skyllas-Kazacos et al.,[6] has great potential for EES compared to other types of RFBs. Since the same vanadium ions with different oxidation states are dissolved in two electrolytes of VRFB, cross-contamination is greatly suppressed. In contrast, such cross-contamination occurs intrinsically in other RFBs, which contain different types of redox species such as catholyte and anolyte.

The main weakness of traditional VRFBs is poor energy density (<25 Wh/L), which results from the low solubility of VO^{2+} (<1.7 M) in the catholyte. In 2011, Li and co-workers[12,13] conducted important research that increased the energy density of a VRFB by up to 70% over the previous conventional system. They found that the solubility of VO^{2+} can increase to 2.5 M in a mixed and highly concentrated acidic solution of H_2SO_4 (2.5 M) mixed with HCl (6 M). This report illustrated the necessity to fully determine the electrochemistry of vanadium species in highly concentrated mixed acidic solution to achieve future high-energy-density VRFBs.

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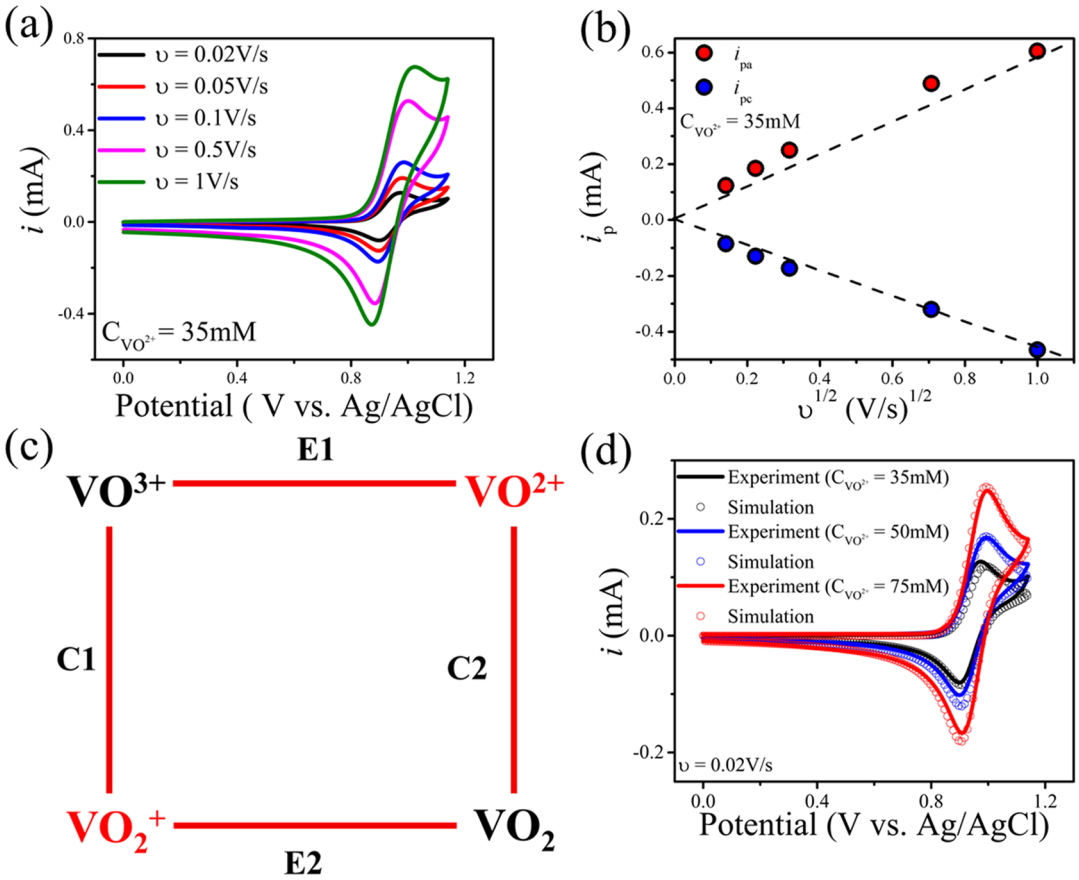

Figure 1. (a) CVs in solutions with $C_{VO^{2+}}$ = 35 mM at different υ values, (b) i_{pa} and i_{pc} vs $\upsilon^{1/2}$, (c) suggested schematic square scheme of the VO^{2+}/VO_2^+ redox reaction, and (d) CVs at υ = 0.02 V/s in solutions containing different $C_{VO^{2+}}$ values and the corresponding best-fit simulation results based on the suggested EC/EC reaction scheme.

For the mechanistic understanding of vanadium redox chemistry, the main half-redox reaction on each cathode and anode in the VRFB is generally known as follows

$$\text{cathode: } VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$$

$$\text{anode: } V^{3+} + e^- \rightleftharpoons V^{2+}$$

Sun and Skyllas-Kazacos reported the reaction mechanism of two redox reactions on the surface of a carbon felt electrode where the mechanism is proposed based on the interaction of a vanadium species with the surface functional group on the carbon electrode. This research utilized detailed surface analyses from X-ray photoelectron spectroscopy (XPS) and scanning electron microscopy (SEM).[14,15] Later, Li et al. reported dependency of electrochemical activities for the V^{3+}/V^{2+} redox reaction on a hydroxyl functional group on the carbon surface.[16]

To better reveal the reaction mechanism from electrochemical analyses, Zhong and Skyllas-Kazacos investigated the kinetics of the VO^{2+}/VO_2^+ redox reaction with different types of carbon electrodes through the rotating disk electrode (RDE) and Tafel analyses.[17] Gattrell et al. also studied the redox reaction of VO^{2+}/VO_2^+ based on the RDE experiments.[18] Liang et al. reported voltammetric analyses of vanadium redox couples on a Vulcan XC72 carbon black electrode.[19] Wu et al. measured the standard rate constants of VO^{2+}/VO_2^+ and V^{3+}/V^{2+} redox reactions on various carbon paper electrodes and revealed that the kinetics of the redox reactions are significantly affected by an electrode material.[20] The proposed mechanism of VO^{2+}/VO_2^+ and V^{3+}/V^{2+} redox reactions on a carbon electrode are well described in respective elementary steps.[21] However, the proposed mechanistic pathway remains questionable because no reports have proved or identified the pathway through voltammetric analyses, such as voltammetric simulation.[21−24]

In addition, the hydrogen evolution reaction (HER) occurs as a side reaction during electroreduction of V^{3+} in a VRFB, which results in the decrease in Coulombic efficiency of a VRFB due to partial consumption of charge during the charging process.[25] Despite the importance of understanding the origin of the parasitic HER occurring in VRFBs, only a few articles have been reported.[26−31] Therefore, the primary reason for this critical side reaction is unknown.

In this article, we present the thorough voltammetric analyses associated with V(V)-V(IV)-V(III)-V(II) redox chemistry occurring in a highly concentrated mixed acidic medium. These analyses are coupled with a voltammetric simulation for understanding the electrochemistry in a high-energy-density VRFB. In addition, we found possible formation of a metastable V(II) species as a side reaction, which could act as an electrocatalyst for the HER. Furthermore, a necessary condition for the existence of this V(II) species is achieved when V^{3+} is electrochemically reduced on a glassy carbon electrode, resulting in a high local concentration of V^{2+} in the vicinity of the electrode surface. This "phantom" effect of this V(II) species can also be systematically explained using the suggested mechanism.

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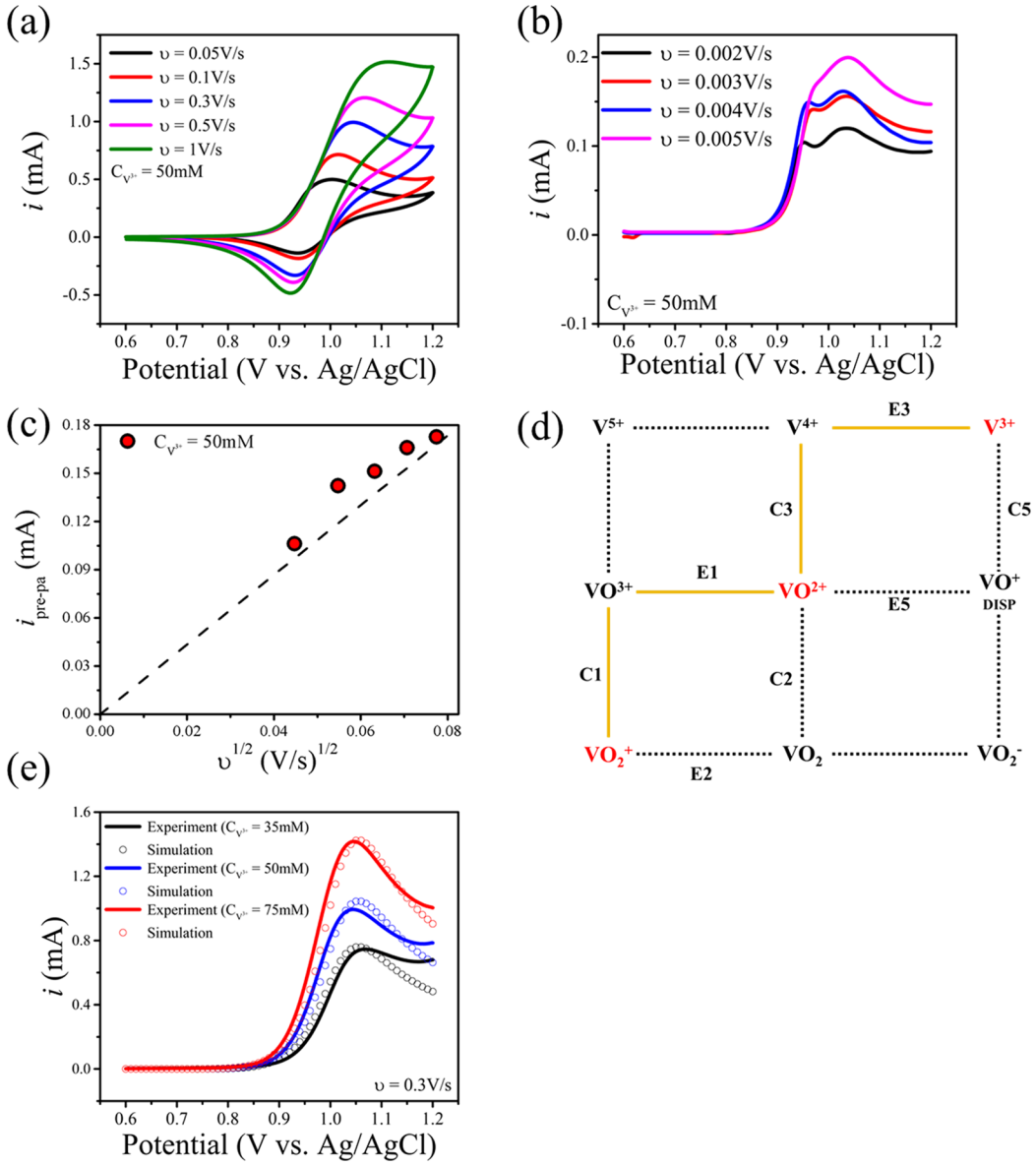

Figure 2. (a) CVs in solution with $C_{V^{3+}}$ = 50 mM at different ν values from 0.05 to 1 V/s. (b) CVs in solution with $C_{V^{3+}}$ = 50 mM at different ν values from 2 to 5 mV/s and (c) i_{pa} from the first oxidation peak attributed to electro-oxidation of V^{3+} to VO^{2+} as a function of $\upsilon^{1/2}$. (d) Schematic description of electro-oxidation of V^{3+} to VO_2^+; the ocher line represents the reaction pathway, and species with red colors are the most stable V species in the corresponding oxidation state. (e) LSVs in solutions with different $C_{V^{3+}}$ values at υ = 0.3 V/s and the corresponding best-fit simulation results based on the suggested ECEC scheme, shown in d.

The presented mechanistic study for the full V(V)-V(IV)-V(III)-V(II) redox reaction and formation of the V(II)O-related vanadium catalyst for the HER during electroreduction of either V^{3+} or VO^{2+} would provide a powerful guideline for understanding electrochemistry in a VRFB and the corresponding parasitic HER during the charge/discharge processes in a battery system.

■ EXPERIMENTAL SECTION

Chemicals. All solutions were prepared with deionized Milli-Q water. Sulfuric acid (H_2SO_4, 98%) and hydrochloric acid (HCl, 98%) were purchased from Daejung Chemicals (Cheongwon, Korea). Vanadium(IV) sulfate oxide hydrate ($VOSO_4 \cdot H_2O$, >98%) was obtained from Sigma-Aldrich.

Instruments and Measurements. A CHI-600e potentiostat (CH Instruments, Austin, TX) was used for all electrochemical measurements. Three electrodes were used in an electrochemical cell for the voltammetric studies: a glassy carbon (GC) macrodisk electrode with a radius of 0.15 cm as a working electrode, Ag/AgCl (1 M KCl) as a reference electrode, and a carbon rod as a counter electrode. For identification of possible deposits on carbon during electroreduction of V^{3+}, a piece of graphite was used as a working electrode. All electrodes were purchased from CH Instruments.

Use of graphite as a counter electrode is particularly important for the study of electroreduction of V(III) and V(IV) species. When a Pt wire was similarly used in other conventional electrochemical cells, a significant amount of Pt migrated from the Pt wire to a working electrode, and the deposited Pt perturbed the electroreduction reaction associated with either the V^{3+}/V^{2+} or VO^{2+}/V^{2+} reaction (see Figure S1).

The electrochemical measurements were conducted in a home-built Faradaic cage, and the glassy carbon was oriented downward. In the experiments for this study, the aqueous solutions were composed of 2.5 M H_2SO_4 and 6 M HCl containing different concentrations of vanadium ions with various oxidation states; when other electrolytes

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are used, it is so noted. All the aqueous solutions were deaerated with Ar for 15 min. The averages of uncompensated resistance (R_s) and double layer capacitance (C_{dl}) of the electrochemical cells in 2.5 M H_2SO_4 + 6 M HCl acidic solutions with some concentration of either $VOSO_4$ or VCl_3 are listed in Table S1.

Simulations of linear sweep voltammograms (LSVs), cyclic voltammograms (CVs), and chronoamperograms (CAs) were performed using DigiElch-Professional v6.F (ElchSoft.com).

Sample morphology and surface composition were observed and measured by field emission scanning electron microscope (FESEM, Verios G4 UC) and energy-dispersive X-ray spectroscopy (EDS, TEAM EDAX, In-TEC Company, USA).

RESULTS AND DISCUSSION

VO^{2+}/VO_2^+ Redox Reaction: EC/EC Square Scheme Mechanism. Figure 1a shows the cyclic voltammograms (CVs) measured in solutions with concentration of VO^{2+}, $C_{VO^{2+}}$, of 35 mM at various scan rates (υ). The ratio of the oxidation peak (i_{pa}) over the paired reduction peak (i_{pc}) is estimated to be ~1, and the potential difference between the oxidation and reduction peaks is 70 mV, which shows that the redox reaction is almost Nernstian. Figure 1b shows i_{pa} and i_{pc} over the square root of scan rates, $\nu^{1/2}$, obtained from CVs at different ν values, which gave linear proportionality with a y intercept of 0. The plots indicate that the VO^{2+}/VO^+ redox reaction is mainly diffusion-controlled, and the outer-sphere reaction model[32] can explain the corresponding electrochemical behavior. From the two obtained plots, the diffusion coefficients of VO^{2+} and VO_2^+ and $D_{VO^{2+}}$ and $D_{VO_2^+}$ were estimated to be 1.1×10^{-6} and 5.7×10^{-7} cm^2/s, respectively. We simulated the CVs based on the suggested square scheme mechanism, as schematically described in Figure 1c; we designated the stable species in the acidic solutions with a red color.

VO^{2+} is electrochemically oxidized to an unstable VO^{3+} intermediate. VO^{3+} is hydrolyzed to form stable VO_2^+ in an acidic condition by liberation of two protons via the EC (electrochemical−chemical) pathway.[11]

$$VO^{2+} \rightarrow VO^{3+} + e^- \quad (1)$$

$$VO^{3+} + H_2O \rightarrow VO_2^+ + 2H^+ \quad (2)$$

In the reverse scan of CV, VO^{2+} is first electrochemically reduced to VO_2. Then, it reacts with two protons and liberates one water molecule to form VO^{2+} through the EC mechanism.

$$VO_2^+ + e^- \rightarrow VO_2 \quad (3)$$

$$VO_2 + 2H^+ \rightarrow VO^{2+} + H_2O \quad (4)$$

Figure 1d shows the CVs for VO^{2+}/VO_2^+ redox reaction in concentrated acidic media containing different $C_{VO^{2+}}$ values at υ = 0.02 V/s and the corresponding best-fit simulation results, which show good agreement with the experimental data. The CVs at different $C_{VO^{2+}}$ values (35, 50, and 75 mM) at υ = 0.05, 0.1, and 0.5 V/s, respectively, are also shown in Figure S2. All of the reactions and corresponding parameters in V(V)-V(IV)-V(III)-V(II) redox chemistry are summarized in Conclusions.

From the electrochemical analysis, we confirmed that the VO^{2+}/VO_2^+ reaction is well explained by the diffusion-controlled outer-sphere reaction model, and there are no other significant factors to perturb the VO^{2+}/VO_2^+ redox reaction.[33−35]

Electro-Oxidation of V^{3+} to VO^{2+}: EC Mechanism. Figure 2a shows CVs in an acidic solution containing $C_{V^{3+}}$ = 50 mM at $0.05 \leq \nu \leq 1$ V/s. The CV at ν = 0.05 V/s, as a representative example, shows one oxidation and reduction peak in the forward and reverse scans, respectively. The i_{pa} of the CV is 1.6 times higher than i_{pc}, indicating that the voltammetric response in the forward scan is associated with electro-oxidation of V^{3+} to VO_2^+ with $2e^-$ transfers, while that of VO_2^+ to VO^{2+} with $1e^-$ transfer occurs in the reverse scan.

At $2 \leq \nu \leq 5$ mV/s, however, a new oxidation peak was observed as a shoulder at 0.95 V, as shown in Figure 2b. One of the possible scenarios is that the pre-peak is attributed to electro-oxidation of strongly adsorbed V^{3+} and $V^{3+}{}_{ads}$ via $2e^-$ transfers to form soluble VO_2^+, and the post-peak is associated with electro-oxidation of diffusional $V^{3+}(aq)$ to VO_2^+ with $2e^-$ transfers. From the unresolved post-peak, diffusion-controlled electro-oxidation of $V^{3+}(aq)$ to VO_2^+ looks seemingly concerted, which actually indicates that the formal potential of V^{3+}/VO^{2+} would be more positive than that of VO^{2+}/VO_2^+;[36] in the opposite case, two sequential voltammetric curves should be obtained. Therefore, in this hypothesis, the onset potential, E_{onset}, of a voltammogram attributed to electro-oxidation of V^{3+} should be positively higher than that of VO^{2+}. However, our experimental results would not follow our hypothesis; E_{onset} for VO^{2+} and V^{3+} were estimated as 0.89 and 0.88 V, respectively. Therefore, the pre-peak is considered to be the resolved peak associated with electro-oxidation of V^{3+} to VO^{2+} via $1e^-$ transfer under a diffusion-controlled condition because i_{pa} from the pre-peak is relatively linear to $\nu^{1/2}$, as shown in Figure 2c. From the voltammogram (Figure 2b), we roughly estimated the formal potential of V^{3+}/VO^{2+} as 0.93 V, which is a modest 0.07 V less positive than that of VO^{2+}/VO_2^+; the small deviation of pre-peak current from linearity to $\nu^{1/2}$ in Figure 2c is mainly because the observed two peaks are not completely separated and thus influence each other.

We simulated a linear sweep voltammogram (LSV) associated with electro-oxidation of V^{3+} to VO_2^+ based on the ECEC scheme where the first EC reactions are as follows, which is also schematically described in Figure 2d

$$V^{3+} \rightarrow V^{4+} + e^- \quad (5)$$

$$V^{4+} + H_2O \rightarrow VO^{2+} + 2H^+ \quad (6)$$

Figure 2e shows the LSVs at 0.3 V/s measured in acidic solutions with different concentrations of V^{3+}, and the corresponding best-fit simulation results are overlaid on the experimental LSVs; LSVs at other υ values in solutions with different $C_{V^{3+}}$ values are shown in Figure S3. The simulation data agree well with the experimental results, indicating that the ECEC pathway for V^{3+} oxidation to VO_2^+ is reasonable. From the simulation, electro-oxidation of V^{3+} to V^{4+} is quasi-reversible, while the VO^{3+}/VO^{2+} redox reaction (reaction 1) is Nernstian, which results in an unresolved oxidation peak at a moderately high υ. $D_{V^{3+}}$ was estimated from the reduction peak associated with V^{3+} reduction to V^{2+}, which is discussed in the next section.

V^{3+}/V^{2+} Redox Reaction from Voltammetric Analysis: E Mechanism. Figure 3a shows CV in solutions with $C_{V^{3+}}$ = 75 mM at $0.05 \leq \nu \leq 0.5$ V/s. The V^{3+}/V^{2+} redox reaction is kinetically slow[8,9,18,20,37,38] and is clearly distinguished from VO^{2+}/VO_2^+; the potential difference between the reduction and oxidation peaks associated with the V^{3+}/V^{2+} redox reaction is about 1.2 V, while that from VO^{2+}/VO_2^+ is only 70 mV. As expected, i_{pc} for V^{3+} reduction to V^{2+} is linearly proportional to

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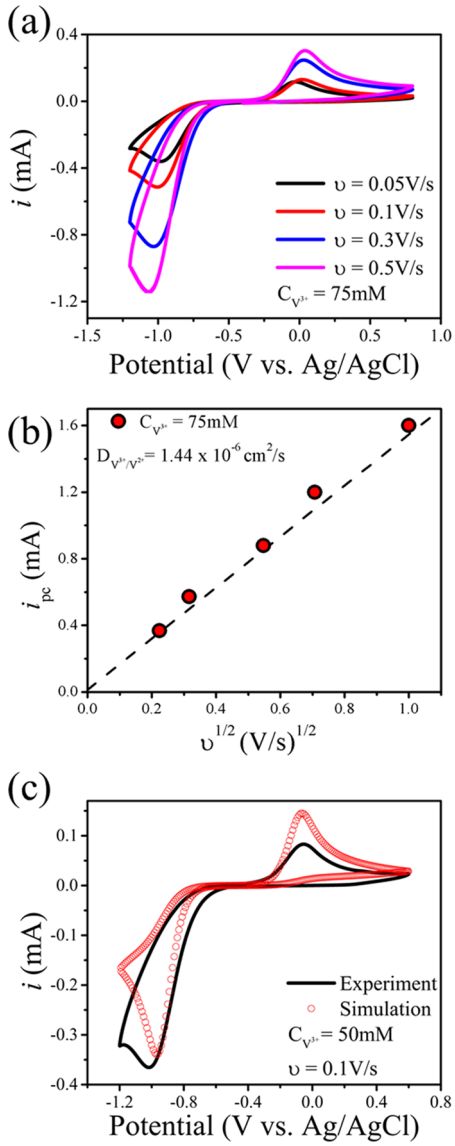

Figure 3. (a) CVs in solution with $C_{V^{3+}}$ = 75 mM at various υ, (b) i_{pc} associated with electroreduction of V^{3+} vs $\upsilon^{1/2}$, and (c) CV in solution containing $C_{V^{3+}}$ = 50 mM at 0.1 V/s and the corresponding best-fit simulation result based on $V^{3+} + e^- \rightleftharpoons V^{2+}$.

$\nu^{1/2}$, as shown in Figure 3b, and $D_{V^{3+}}$ is estimated to be 1.44 × 10^{-6} cm^2/s.

Figure 3c shows CVs measured in solution containing $C_{V^{3+}}$ = 50 mM at ν = 0.1 V/s, and the corresponding best-fit simulation results are provided as a representative example; other CVs and the corresponding simulation results measured in different $C_{V^{3+}}$ values at various ν are shown in Figure S4. It is well known that the most thermodynamically stable V(III) and V(II) species are V^{3+} and V^{2+} in an acidic aqueous solution, respectively.[39] Therefore, the simulation was conducted under the simple 1e$^-$ transfer reaction

$$V^{3+} + e^- \rightleftharpoons V^{2+} \quad (7)$$

The simulation results indicate that the V^{3+}/V^{2+} redox reaction is electrochemically irreversible on a GC electrode.[11,18]

Formation of a Heterogeneous Electrocatalyst for the Hydrogen Evolution Reaction (HER) during Electroreduction of V^{3+}. We found a side electrode reaction occurring in electroreduction of V^{3+},[40] the parasitic hydrogen evolution reaction (HER).[25]

Figure 4a shows the 1st and 20th cycle CVs, driven by continuous potential cycling, with 20 cycles in acidic solution

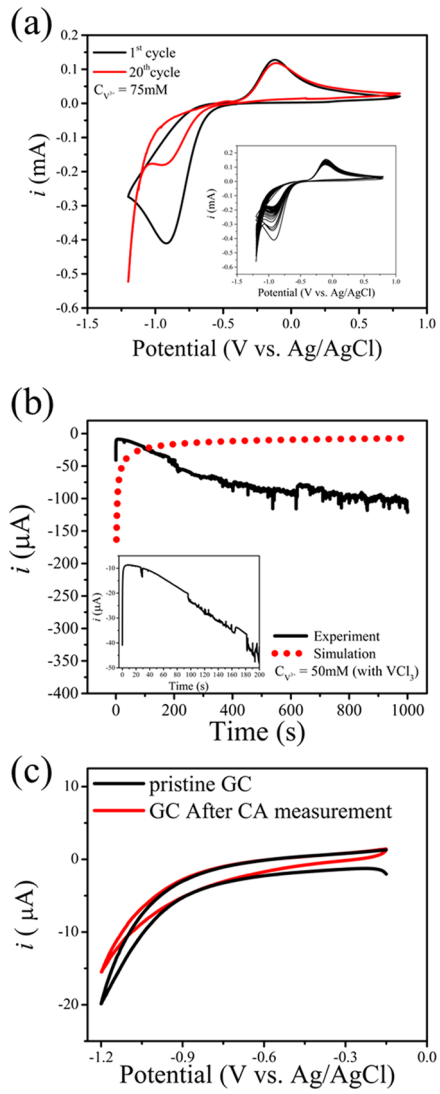

Figure 4. (a) First (black) and 20th (red) cycle CVs in solution containing $C_{V^{3+}}$ = 75 mM from the successive CVs shown in the inset of a. (b) CA (black line) at the constantly applied −1.1 V potential for 1000 s in solution containing $C_{V^{3+}}$ = 50 mM and the theoretical Cottrell decay curve (red dots) associated with the 1e$^-$ transfer reduction process: (inset) the enlarged graph to better observe the small current region. (c) CVs in solution without V^{3+} to observe the HER on a pristine (black) and GC (red) electrode after electrochemical reduction of V^{3+} in solution with $C_{V^{3+}}$ = 50 mM.

containing $C_{V^{3+}}$ = 75 mM; the entire set of successive CVs is shown in the inset of Figure 4a. As the number of CV cycles increased, both i_{pc} and i_{pa} from the V^{3+}/V^{2+} redox reaction decreased. This electrochemical characteristic indicates that the electrode site for the V^{3+}/V^{2+} redox reaction was gradually blocked by an adsorbed species that is electrochemically inactive in this potential region. Second, we did not observe any significant reduction current associated with the HER in the first cycle CV. However, as the number of CV cycles increased, the overpotential of HER decreases, and eventually, the reduction current from the HER was significant in the 20th

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cycle CV. From the above results, we could conclude that an adsorbed V(II) species blocks electroreduction of diffusional V^{3+}, while it electrocatalyzes the HER. Henceforth, we designated this adsorbed V(II) species as $V(II)_{catalyst}$.

Formation of $V(II)_{catalyst}$ is clearly observed in a chronoamperometric $i-t$ curve. Figure 4b shows the chronoamperogram (CA) measured in solution containing 50 mM V^{3+} at a constantly applied potential of −1.1 V for 1000 s. After the sharp current drop and until t reaches 60 s, the reduction current is continuously increased (see the inset of Figure 4b). More interestingly, as t passed beyond 200 s, numerous stochastic current spikes were observed with a significant increase in overall reduction current. There are no chances for stochastic current spikes due to the collision of electrocatalytic particles in the HER, such as Pt. As we used a carbon rod as the counter electrode in the electrochemical cell, it is not possible for Pt particles to migrate from the counter to working electrode. Recently, Dick and Bard reported the deciphering of stochastic current spikes, which are associated with the HER by electrochemically formed discrete metal clusters.[41] In our condition, we believe that the individual current spikes might signify formation of $V(II)_{catalyst}$ clusters during electroreduction of V^{3+}. However, we consider the analysis of the observed current spikes and possible formation of discrete $V(II)_{catalyst}$ clusters beyond our scope and focus on revealing the reaction mechanism during the redox reaction of vanadium species.

Since electroreduction of V^{3+} to V^{2+} is a diffusion-controlled $1e^-$ transfer reaction, it should demonstrate Cottrell behavior, which is shown in Figure 4b as a red dotted curve under the condition of no perturbation of the V^{3+}/V^{2+} redox reaction. Until t reached 6 s, the reduction current from the experimental measurement was smaller than the theoretical diffusion-controlled $1e^-$ transfer reduction current due to partial blockage of the diffusional V^{3+} reduction reaction to V^{2+} by $V(II)_{catalyst}$. However, as the experimentally obtained reduction current became significantly higher than the theoretical current after 102 s and the most probable reaction to perturb electroreduction of V^{3+} is the HER in our condition, the oxygen reduction reaction (ORR) can be excluded because we vigorously purged the solution with Ar before the experiment.

The observed electrochemical behavior was only observed during electroreduction of V^{3+} in acidic solution. On a pristine GC electrode, we confirmed that such a dramatic reduction in current from the HER at that constant potential did not occur over the same time window (see Figure S5). Also, we confirmed that this increasing current is "not" attributed to redox cycling of V^{3+} to V^{2+} due to a possible galvanic exchange reaction between V^{2+} and VO_2^+, which is produced on a counter electrode. If this galvanic exchange reaction really occurred in the electrochemical cell, we should also observe the catalytic current over time when V^{3+} is electrochemically oxidized to VO_2^+ on a working electrode. Figure S6 shows a CA curve at a constantly applied potential of 1.1 V to drive electro-oxidation of V^{3+} to VO_2^+ in solution with $C_{V^{3+}}$ = 50 mM. The observed $i-t$ curve following a Cottrell decay indicates that the possible catalytic current amplification by the galvanic exchange reaction did not occur in our condition.

After chronoamperometric measurement in V^{3+} solution, as shown in Figure 4b, we removed the GC electrode from the electrochemical cell, carefully rinsed it, and used it as the working electrode in solution without V^{3+}. The CV curve from the GC electrode shows almost the same voltammetric response for the HER as that from a pristine GC electrode, as shown in Figure 4c, indicating that $V(II)_{catalyst}$ only exists in the vicinity of a GC electrode when the local concentration of V^{2+} at the surface, $C_{V^{2+},loc}$, is high. To verify metastable characteristics of $V(II)_{catalyst}$ during electroreduction of V^{3+} on a GC electrode, ex situ SEM and EDS measurements were performed after electroreduction of V^{3+} to V^{2+} on a graphite rod electrode as the working electrode in solution containing $C_{V^{3+}}$ = 50 mM. In this reaction, we did not observe any detectable precipitated V species on a carbon surface, as shown in Figure S7a; the SEM images and the corresponding EDS data from a pristine electrode and graphite rod used as a working electrode for electroreduction of VO^{2+} in acidic solution containing $C_{VO^{2+}}$ = 50 mM are shown in Figure S7b,c.

We believe that $V(II)_{catalyst}$ is strongly related to V(II)O on a GC electrode and is conditionally formed based on the following proposed mechanism. For V(II) species, V^{2+} could hydrolyze to VO with liberation of $2H^+$.

$$V^{2+} + H_2O \rightleftharpoons VO + 2H^+ \quad (8)$$

The equilibrium of the above hydrolysis reaction is not thermodynamically preferred because V^{2+} is the most stable V(II) species in an acidic condition. However, V^{2+} could kinetically undergo a transition to VO only under the condition in which $C_{V^{2+},loc}$ is high enough to render the forward rate faster than the backward. At completion of electroreduction of V^{3+}, VO immediately reacts with H^+ and returns to soluble V^{2+}, resulting in the absence of $V(II)_{catalyst}$ on a GC surface after a chronoamperometric measurement. The electrocatalysts, which can only conditionally exist, were also reported elsewhere.[42−45] Figure 5 shows the summarized scheme of the mechanistic pathway for electroreduction of V^{3+}.[25−27,29,46]

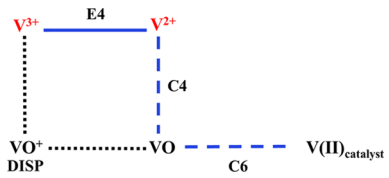

Figure 5. Schematic description of the electroreduction of V^{3+} to V(II). The blue line represents the reduction pathway, and the reaction passage through the blue dotted line is only possible under the condition in which the forward reaction of C4 toward VO is kinetically favored.

Although we observed a significant reduction current over time due to the HER by the possibly formed $V(II)_{catalyst}$, the variance of the reduction current associated with the HER is quite large, as shown in Figure S8. This indicates that formation of $V(II)_{catalyst}$ is also dependent on the condition of a GC surface, which is difficult to control in a study.[14−16,20,21,47]

Electroreduction of VO^{2+} to V(II): EEC-DISP Mechanism and HER by $V(II)_{catalyst}$. We obtained more electrochemical information of $V(II)_{catalyst}$ based on analysis of VO^{2+} reduction to V(II). Figure 6a−c shows CVs measured in solution with different $C_{VO^{2+}}$ values (35, 50, and 75 mM) at ν = 0.05 V/s. In the first segment of the CV at $C_{VO^{2+}}$ = 75 mM, for example, the potential was negatively swept from 0 to −1.6 V. We only observed one reduction wave with i_{pc} = −0.6 mA. From i_{pc} and $D_{VO^{2+}} = 1.1 \times 10^{-6}$ cm^2/s, we found the reduction

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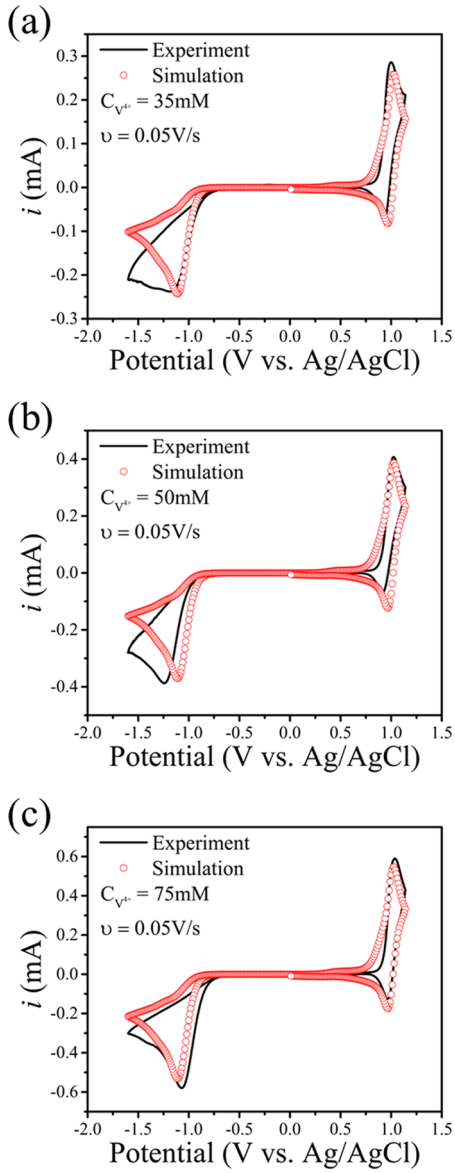

Figure 6. CVs of (a) 35 mM, (b) 50 mM, and (c) 75 mM VO^{2+} at 0.05 V/s and the corresponding best-fit simulation result based on the EEC-DISP (electroreduction of VO^{2+} to V(II) described in Figure 9) + EC/EC (VO^{2+}/VO_2^+ redox reaction described in Figure 1c) scheme.

wave to be associated with electroreduction of VO^{2+} to V(II) species with $2e^-$ transfers. Also, we found the E_{onset} of the reduction peak to be −0.2 V more negative than that measured in V^{3+} solution.

In the second segment of the CV, the potential was positively biased to −1.6 V. Surprisingly, there is no peak associated with electro-oxidation of V^{2+} to V^{3+}. The absence of the oxidation peak indicates that the main V(II) species, which is reduced from VO^{2+} and exists in the diffusion layer on a GC electrode within the given time, is not V^{2+}. Lastly, the oxidation wave attributed to electro-oxidation of VO^{2+} shows a higher i_{pa} (= 0.59 mA) than that from electro-oxidation of VO^{2+} to VO_2^+ as the first segment step in the CV, which has a lower current of 0.36 mA (see Figure 1d).

Based on the above experimental results, we proposed the following mechanism for electroreduction of VO^{2+}. At first, VO^{2+} is electrochemically reduced to VO^+.

$$VO^{2+} + e^- \rightarrow VO^+ \quad (9)$$

Because VO^+ is an unstable intermediate V(III) species, it either reacts with $2H^+$ to form V^{3+} or is electrochemically reduced to VO. If the former case was the main reaction pathway, most of the VO^+ would change to V^{3+} that is finally reduced to the stable V^{2+} species, as stated in the previous section; we note that $V(II)_{catalyst}$ could only form when the time for electroreduction of V^{3+} was sufficiently long enough to render a high $C_{V^{2+},\,loc}$.

Based on the experimental results, we can exclude this pathway because there is no peak from electro-oxidation of V^{2+} in the CVs (Figure 6). This indicates that the following reaction with $2H^+$ transfers is too kinetically slow to form V^{3+} in our time scale of cyclic voltammetry, which is about 0.50 s (= $RT/F\upsilon$) in spite of the spontaneity of the reaction. This time, it is found with a gas constant R of 8.31 J/(mol·K), T as the temperature of 25 °C, and F as the Faraday constant of 96,485 C/mol. As an alternative, we consider that VO^+ is electrochemically reduced to VO and two VO^+ are disproportionate to VO^{2+} and VO. Finally, as stated in electroreduction of V^{3+}, VO forms $V(II)_{catalyst}$ on a GC electrode surface.

$$VO^+ + e^- \rightarrow VO \quad (10)$$

$$2VO^+ \rightleftharpoons VO + VO^{2+} \quad (11)$$

$$VO \rightarrow V(II)_{catalyst} \quad (12)$$

In a manner similar to this VO^+ reaction, the possible H^+ transfer reaction of VO to V^{2+} is also considered kinetically slow. From the proposed mechanism, as also described in Figure 7, we fitted the simulated CVs (red hollow dots) for the

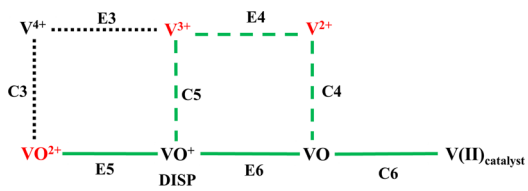

Figure 7. Schematic description of electroreduction of VO^{2+} to V(II). The green line represents the reduction pathway, and the reaction passage through the green dotted line is only possible where the forward reaction of C4 toward VO is kinetically favored.

electroreduction of VO^{2+}, which agreed well with the experimental results in Figure 6, indicating that the suggested EEC-DISP mechanism is reasonable.

We found it difficult to obtain well-defined CVs as υ increased, as described in Figure S9. This ill-defined voltammetric response at higher υ indicates electroreduction of VO^{2+}, which is not fully explained by the suggested mechanism based on the outer-sphere reaction model, and the reaction would be more complicated, including the interaction between the intermediate species and a GC surface, which we did not take account in our simulation.[48]

The formation of a $V(II)_{catalyst}$ during electroreduction of VO^{2+} was more clearly observed when cyclic voltammetry is successively performed to drive electroreduction of VO^{2+}, as shown in Figure 8a. The diffusional electroreduction of VO^{2+} was blocked while the HER was electrocatalyzed due to formation of $V(II)_{catalyst}$, which is similarly driven by V^{3+} reduction.

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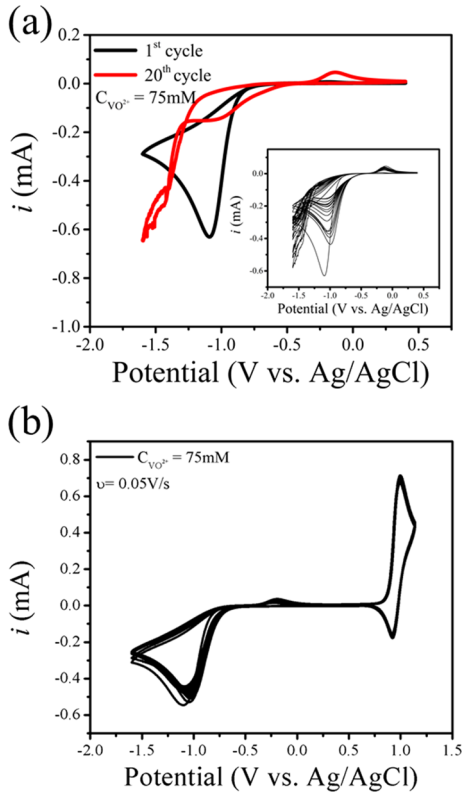

Figure 8. (a) First (black) and 20th (red) cycle CVs in solution containing $C_{VO^{2+}}$ = 75 mM from the successive CVs shown in the inset of a; for obtaining one-cycle CV, the potential was initiated at 0.4 V and negatively swept to −1.6 V. Then, the potential returned to the original point. (b) Successive 20 cycle CVs in solution containing $C_{VO^{2+}}$ = 75 mM; for obtaining the one-cycle CV, the potential was started at 0 V and negatively swept to −1.6 V. Then, the potential was positively swept to 1.2 V and back to the original point.

Interestingly, at the first cycle CV in the successive CVs of Figure 8a, as mentioned earlier, no oxidation peak associated with electroreduction of V^{2+} to V^{3+} was shown. However, we can clearly observe the corresponding oxidation peak in the reverse scan of the 20th cycle CV; the corresponding oxidation peak was gradually increased until the number of CV cycles reached 20, as shown in the inset of Figure 8a. The successive CVs indicate that the fraction of V^{2+} among V(II) species somewhat increased and was associated with the accelerated forward kinetics of VO^{+} to V^{3+} as the number of CV cycles increased. This observation also infers that the chemical reaction of VO^{+} to V^{3+} with $2H^{+}$ transfers is a complicated inner-sphere reaction that is sensitive to a GC surface, although, here, we consider that all chemical reactions during vanadium electrode reactions are homogeneous for the sake of simplicity.[22]

When cyclic voltammetry was successively cycled in a wider potential range from −1.6 to 1.14 V, as shown in Figure 8b, the peak from VO^{2+} reduction did not decrease, and the reduction current associated with the HER was not shown. Although the number of CV cycles increased, this observation results from dissipation of $V(II)_{catalyst}$ due to kinetic acceleration toward VO in reaction 12, which is induced by depletion of VO in the vicinity of a GC electrode based on the suggested mechanism.

Computational Study of the Hydrogen Evolution Reaction (HER) Electrocatalyzed by Vanadium and Vanadium Oxide Catalyst, $[V(H_2O)_6]^{p+}$ and $[VO(H_2O)_5]^{q+}$ with p = 2 and 3 and q = 0, 1, and 2. The B3LYP functional[49,50] combined with the 6-31+G(d,p) basis set[51] was employed to investigate the catalytic properties of V^{3+}, V^{2+}, V(II)O, V(III)O^{+}, and V(IV)O^{2+} toward the hydrogen evolution reaction (HER) in acidic media. For initial geometries of solvated vanadium cations and those of the vanadium oxide species, either six or five water molecules are coordinated to vanadium cations (V^{p+} with p = 2 and 3) and oxide vanadium species (VO^{q+} with q = 0, 1, and 2), respectively, with tetrahedral symmetry, as illustrated in Figure S10.

To mimic the acidic media, H^{+} is bound to oxygen, labeled O_2 (Figure S10c), of the V−OH_2 and V═O bonds in $[V(H_2O)_6]^{p+}$ and $[VO(H_2O)_5]^{q+}$ (p = 2 and 3 and q = 0, 1, and 2). These initial geometries were first optimized using B3LYP/6-31+G(d,p) and followed by calculations of vibrational frequencies at the same theory level to ensure that minima were reached. The optimized geometrical parameters of the equilibrium geometries of all complexes studied in the present work are listed in Table S2.

The following mechanism for the HER was used to evaluate the catalytic properties of the vanadium species

$$\text{catalyst} + H^{+} \rightarrow \text{catalyst} - H^{+} \quad (13)$$

$$\text{catalyst} - H^{+} + e^{-} \rightarrow \text{catalyst} - H \quad (14)$$

$$\text{catalyst} - H \rightarrow \text{catalyst} + 1/2H_2 \quad (15)$$

where catalyst stands for $[VO(H_2O)_5]^{2+}$, $[VO(H_2O)_5]^{+}$, or $[VO(H_2O)_5]$. The same mechanism was used recently by Chen et al.[52] in their studies on transition metal-based electrocatalysts for the HER and by Zhang et al.[53] in their investigation of catalytic activities of molybdenum phosphide in the HER.

The data in Table S2 shows that the V−O bond distances in $[V(H_2O)_6]^{p+}$ and $[VO(H_2O)_5]^{q+}$ are elongated compared to their homologous protonated species, unlike the V═O bonds, which appear to be longer in acidic media. The mean bonds of these protonated vanadium and vanadium oxide species occur in the same manner as the experimental mean bond distances of $[V(H_2O)_6]^{3+}$ and $[VO(H_2O)_5]^{2+}$, which were measured in

Table 1. Relative Energies of Vanadium Oxide-Based Catalysts for the Three Steps (Reactions 13−15) of the HER Computed Using B3LYP/6-31+G(d,p)[a]

complex	sym	LF	state	PA (kcal/mol)	EA (kcal/mol)	FE (kcal/mol)
$[VO(H_2O)_5]^{2+}$	C_{2v}	109	$^2A_1$	−7.53	364.95	−12.76
$[VO(H_2O)_5]^{+}$	C_s	18	$^1A'$	141.90	252.14	23.86
$[VO(H_2O)_5]$	C_1	58	^2A	308.11	31.213	−30.86

[a]PA, EA, and FE stand for proton affinity, electron affinity, and formation energy of the hydrogen molecule, respectively.

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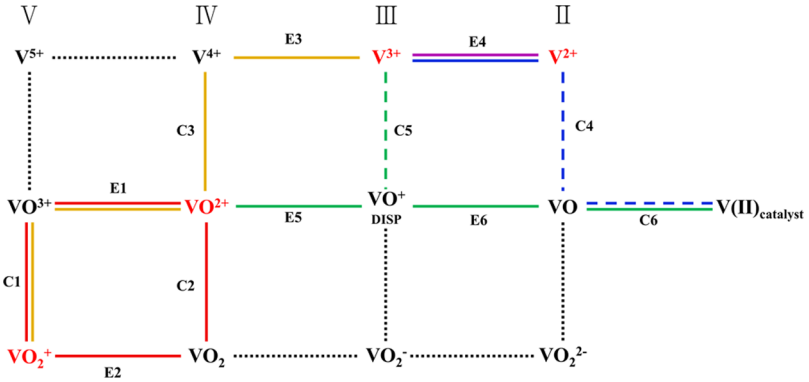

Figure 9. Overall reaction mechanism of the V(V)-V(IV)-V(III)-V(II) redox reaction in 2.5 M H_2SO_4 + 6 M HCl aqueous solution. The species with a red color represents the most stable in the corresponding oxidation state: redox reaction of VO^{2+}/VO_2^+, $E_1C_1/E_2/C_2$ (red solid line); electro-oxidation of V^{3+} to VO_2^+, $E_3C_3E_1C_1$ (ocher solid line); redox reaction of V^{3+}/V^{2+} in a short time window ($\tau \leq 1.3$ s), E_4 (blue solid line); electroreduction of V^{3+} in a second-long time window ($1.3\ s \leq \tau$), $E_4C_4C_6$ (blue solid + dotted lines); and electroreduction of VO^{2+}, $E_5E_6C_6$-DISP (green solid line) including the side reaction over CV cycles, C_5 (green dotted line).

Table 2. All of the Reactions and Corresponding Parameters in Simulations Based on the Suggested Mechanism[a]

reaction and parameters for CV simulation on GC		
E1	$VO^{3+} + e^- \rightleftharpoons VO^{2+}$	$E_{1°} = 1.03$ V, $\alpha = 0.5$, $k_{et} = 46$ cm/s
C1	$VO^{3+} + H_2O \rightleftharpoons VO_2^+ + 2H^+$	$K_{eq} = 28\ M^2$, $k_f = 50\ s^{-1}$, $k_b = 1.8\ M^{-2}\ s^{-1}$
E2	$VO^{3+} + e^- \rightleftharpoons VO^{2+}$	$E_{2°} = 0.75$ V, $\alpha = 0.5$, $k_{et} = 0.08$ cm/s
C2	$VO_2 + 2H^+ \rightleftharpoons VO^{2+} + H_2O$	$K_{eq} = 1.44 \times 10^3\ M^{-2}$, $k_f = 1 \times 10^7\ M^{-2}\ s^{-1}$, $k_b = 7.0 \times 10^3\ s^{-1}$
E3	$V^{3+} + e^- \rightleftharpoons V^{4+}$	$E_{3°} = 0.96$ V, $\alpha = 0.5$, $k_{et} = 0.0018$ cm/s
C3	$V^{4+} + H_2O \rightleftharpoons VO^{2+} + 2H^+$	$K_{eq} = 1 \times 10^7\ M^2$, $k_f = 20\ s^{-1}$, $k_b = 2 \times 10^{-6}\ M^{-2}\ s^{-1}$
E4	$V^{3+} + e^- \rightleftharpoons V^{2+}$	$E_{4°} = -0.51$ V, $\alpha = 0.5$, $k_{et} = 5 \times 10^{-7}$ cm/s
E5	$VO^{2+} + e^- \rightleftharpoons VO^+$	$E_{5°} = -0.27$ V, $\alpha = 0.5$, $k_{et} = 1.27 \times 10^{-9}$ cm/s
E6	$VO^+ + e^- \rightleftharpoons VO$	$E_{6°} = -0.15$ V, $\alpha = 0.5$, $k_{et} = 3 \times 10^{-12}$ cm/s
C6	$VO \rightleftharpoons VO_{cat}$	$K_{eq} = 8.3$, $k_f = 1.7 \times 10^9\ s^{-1}$, $k_b = 1.5 \times 10^8\ s^{-1}$
DISP	$VO^+ + VO^+ \rightleftharpoons VO^{2+} + VO$	$K_{eq} = 96$, $k_f = 7.5 \times 10^2\ s^{-1}$, $k_b = 7.86\ s^{-1}$

[a]Shown in Figure 9; in the digital simulation, neither C4 nor C5 is included.

water containing $HClO_4$ by Krakowiak et al.[54] and by Hodson et al.[55]

The optimized geometries of the protonated vanadium catalysts $[V(H_2O)_6H]^{3+}$ and $[V(H_2O)_6H]^{4+}$ were unstable and tended to dissociate in hydronium ions, $[V(H_2O)_5]^{2+}$, and $[V(H_2O)_5]^{3+}$ (Table S2 and Figure S11a). Stable species were obtained when the proton was attached to oxygen in $[VO(H_2O)_5]$, $[VO(H_2O)_5]^+$, and $[VO(H_2O)_5]$ (Figure S11b, Table 1, and Table S2).

The relative energies of the neutral and cation vanadium oxide for the HER (reactions 13−15) are presented in Table 1. According to the data, solvated V(II)═O has a higher proton affinity (PA = 308.11 kcal/mol) than $V(III)O^+$ and $V(IV)O^{2+}$. The electron affinity calculations (reaction 14) reveal that all the protonated oxide vanadium species studied in the present paper are in a favorable state to receive one more electron. The electron affinities decrease in the following order: $[VO(H_2O)_5H]^{3+}$ (365 kcal/mol) > $[VO(H_2O)_5H]^{2+}$ (252 kcal/mol) > $[VO(H_2O)_5H]^+$ (31 kcal/mol). After receiving one electron (reaction 14), the formed vanadium complexes ($[VO(H_2O)_5H]^{q+}$, with q = 0, 1, and 2) release a hydrogen molecule (reaction 15). Neutral vanadium oxide, $[VO(H_2O)_5]$, is identified in Table 1 to possess the most favorable formation energy of the hydrogen molecules, estimated to be about −30.86 kcal/mol, followed by $[VO(H_2O)_5]^+$ (−12.76 kcal/mol). The computed hydrogen formation energy from $[VO(H_2O)_5]^{2+}$ is unfavorable (23.86 kcal/mol). Analogously, the low-valent hydroxide cation $V{-}O{-}H^+$ was found, by Fourier transform ion cyclotron resonance (FT-ICR) mass spectrometry, to play an important role in gas-phase dehydrogenation of methanol.[56] Therefore, oxide vanadium, V(II)═O, appears to be the most plausible electrocatalyst for the HER among the vanadium cations and vanadium oxide species studied in the present work.

■ CONCLUSIONS: OVERALL ELECTROCHEMICAL V(V)-V(IV)-V(III)-V(II) MECHANISM AND ELECTROCATALYTIC HER BY $V(II)_{CATALYST}$ INDUCED BY ELECTROREDUCTION OF V(III)

In summary, the total schematic reaction map of charge transfers and water proton transfers in V(V)-V(IV)-V(III)-V(II) species are shown in Figure 9. This map can guide each electrode and chemical reaction, and the parameters of the corresponding reactions are listed in Table 2.

The most stable V species at different oxidation states, VO_2^+, VO^{2+}, V^{3+}, and V^{2+}, are marked with red color in Figure 9. The $V(IV)O^{2+}/V(V)O_2^+$ redox reaction is easily estimated to follow an E_1C_1/E_2C_2 square scheme reaction mechanism (red solid line). For electro-oxidation of V^{3+} to VO^{2+}, the $E_3C_3E_1C_1$ pathway is reasonable: $V^{3+} \rightarrow V^{4+} \rightarrow V(IV)O^{2+} \rightarrow V(V)O^{3+} \rightarrow V(V)O^{2+}$ (the solid ocher line).

The V^{3+}/V^{2+} redox reaction in a short time window ($\tau \leq 1.3$ s, which is estimated from the characteristic time of CV at 0.02 V/s)[32] is electrochemically irreversible on a GC electrode (E_4), and during electroreduction of V^{3+} in a relatively long time

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window ($1.3 \leq \tau$, which is estimated from the CA in Figure 4b), $V(II)_{catalyst}$ for the HER formed when the fraction of VO was kinetically increased with $C_{V^{2+},loc}$ (blue dotted line), following the $E_4C_4C_6$ pathway. Possible formation of $V(II)_{catalyst}$, which is related to the V(II)O species, and its electrocatalytic property for the HER is strongly supported by the computational calculation results.

The electroreduction pathway of $V(IV)O^{2+}$ is significantly different from that of V^{3+}. We found $V(IV)O^{2+}$ to be reduced to VO and $V(II)_{catalyst}$ to form eventually based on the $E_5E_6C_6$-DISP pathway (green solid line). In addition, we confirmed that V^{3+} formation was small during $V(IV)O^{2+}$ reduction, probably due to activation of a GC surface for kinetic acceleration toward V^{3+} in reaction C5 (see Figure 9) as the number of CV cycles increases (green dotted line).

Although all chemical reactions involving in the suggested mechanism are considered homogeneous, we believe that the corresponding reactions would be more complicated due to a nature of heterogeneity involving a carbon surface as a reaction site, and functionality of a carbon surface would strongly influence on kinetics of the chemical reactions in reality. Carbon felts are practically used as electrodes in the VRFB, which would have the conditions with different surface functionalities compared to a GC. However, we consider that kinetics of the chemical reactions would be changed depending on the types of carbon electrodes, while the main reaction framework of the vanadium redox mechanism would be similar.

We hope that this reaction map could help to design the surface of carbon electrodes in the VRFB through various modification approaches to enhance electrochemical activities of vanadium redox reactions and inhibit the parasitic HER.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.9b12676.

> Measured average uncompensated resistance and double layer capacitance in 6 M H_2SO_4 + 2.5 M HCl with various concentrations of either $VOSO_4$ or VCl_3, more voltammetric and chronoamperometric data, geometric models, and corresponding parameters for computational calculations (PDF)

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Notes

The authors declare no competing financial interest.

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